Ben Robles

Senior Software Engineer

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Accomplished Senior Software Engineer with over a decade of experience architecting, developing, and optimizing backend systems, distributed microservices, and cloud-native infrastructure across Netflix, AWS, and Riot Games. Expert in designing scalable, low-latency, fault-tolerant backend platforms with deep hands-on expertise in Java, Kotlin, Go, Python, and AWS-native services. Proven record of building automation frameworks, network orchestration systems, and resilient service meshes serving hundreds of millions of users globally. Adept in system design, performance optimization, observability, and CI/CD automation, combining strong software craftsmanship with cross-functional collaboration and leadership.

SKILLS

- ✓ **Programming Languages:** Java, Kotlin, Go (Golang), Python, C, C++, Bash, Shell Scripting, JavaScript (Node.js), TypeScript
- ✓ **Frameworks & Libraries:** Spring Boot, Micronaut, Quarkus, Flask, FastAPI, gRPC, REST, GraphQL, OpenAPI, JUnit, Mockito
- ✓ **Cloud Platforms:** AWS (EC2, ECS, EKS, Lambda, DynamoDB, S3, CloudFormation, CloudWatch, CDK, RDS, Route53, API Gateway, SQS, SNS, Step Functions, ElastiCache, QLDB), Azure, GCP
- ✓ **Infrastructure & DevOps:** Docker, Kubernetes, Helm, Terraform, Ansible, Jenkins, GitHub Actions, TeamCity, ArgoCD, Spinnaker, Prometheus, Grafana, ELK Stack, Datadog, New Relic
- ✓ **Architecture & Backend Design:** Microservices, Event-Driven Architecture, CQRS, Domain-Driven Design (DDD), Pub/Sub, Message Queues, Service Mesh, API Gateway, Caching Strategies, Distributed Systems, Horizontal Scaling, High Availability (HA), Load Balancing, Resilience4j
- ✓ **Databases & Data Engineering:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, ElasticSearch, Kafka, Kinesis, RabbitMQ, Snowflake, Cassandra, Aurora, Presto, Airflow, dbt
- ✓ **Testing & Quality:** Unit/Integration Testing, Load Testing (JMeter, Locust), Chaos Engineering, Canary Deployments, Blue-Green Deployments
- ✓ **Security & Networking:** OAuth2, JWT, IAM, TLS, VPNs, TCP/IP, UDP, Firewalls, Packet Inspection, DDoS Protection, Zero Trust, SSO
- ✓ **System Design Expertise:** CAP Theorem, Consistency Models, Eventual Consistency, Data Replication, Sharding, Caching, API Rate Limiting, Circuit Breakers
- ✓ **Tools & Collaboration:** Jira, Confluence, Git, Bitbucket, Agile/Scrum, Pair Programming, Code Review, CI/CD Pipelines

WORK EXPERIENCE

Netflix - Senior Software Engineer (Remote)
Apr 2022 - Present

- Lead backend engineer within the Open Connect infrastructure team, building distributed backend systems that manage Netflix's global CDN platform serving 250M+ users and petabytes of video data daily.
- Architected microservices in Go and Kotlin for orchestrating CDN node deployment, traffic routing, and global caching, improving operational automation throughput by 70%.
- Built control-plane APIs integrating with AWS ECS, Lambda, DynamoDB, and CloudWatch to automate health checks, scaling events, and failover responses with near-zero latency.
- Designed intelligent traffic-balancing logic using gRPC and service discovery patterns, reducing packet loss and regional congestion by 35%.
- Integrated Prometheus + Grafana dashboards for distributed observability and proactive anomaly detection, enabling real-time traffic insights across 1,000+ global edge sites.
- Implemented zero-downtime blue/green deployments using Spinnaker, Jenkins, and Helm, achieving 99.999% uptime for Open Connect's API ecosystem.
- Built internal developer tooling to simplify CI/CD for backend services — reducing developer onboarding time by 40%.
- Partnered with data analytics and video encoding teams to align infrastructure metrics with business KPIs on streaming performance and user experience.

AWS - Software Engineer
Dec 2019 - Apr 2022

- Delivered large-scale distributed backend systems powering AWS's private connectivity (DirectConnect) and internal automation tools for global finance teams.
- Designed fault-tolerant backend services in Java, Kotlin, and Go, enabling multi-region hybrid connectivity with sub-second latency for Fortune 500 customers.
- Engineered data-plane orchestration logic across ECS, Lambda, and API Gateway, handling millions of provisioning requests per week.
- Built asynchronous processing pipelines using SQS, SNS, Step Functions, and Kinesis, achieving horizontal scalability and backpressure resilience.
- Deployed DynamoDB- and QLDB-backed transactional systems ensuring strong consistency for connection metadata and billing records.
- Integrated CI/CD workflows using Jenkins and CDK to support automated testing, rollback, and version control across 50+ services.

- In the Finance Automation group, led backend design of an enterprise microservices platform for internal accounting teams using Java + GraphQL + ElasticSearch, improving process efficiency by 35%.
- Enhanced backend observability by implementing distributed tracing with AWS X-Ray and centralized logging via CloudWatch + ELK.
- Introduced cost-optimization policies through automation scripts, reducing idle resource spend by 22% across dev environments.
- Conducted cross-team performance deep dives on service throttling, optimizing API throughput under high concurrency.

Riot Games - Software Engineer

Jun 2017 - Dec 2019

- Led backend and network automation engineering for Riot Direct, the proprietary network serving millions of players in League of Legends and Valorant.
- Built network device automation platform in Python + Nornir + Napalm + Jenkins, enabling safe global configuration rollouts across thousands of routers and switches.
- Developed distributed orchestration services in Go and C to coordinate real-time traffic inspection, latency monitoring, and DDoS mitigation.
- Introduced in-house packet inspection system using eBPF/XDP, saving Riot millions annually versus vendor solutions.
- Implemented microservice-based orchestration layer for inspection servers, leveraging REST APIs, gRPC, and Redis caching for state synchronization.
- Designed configuration versioning and rollback system integrated with Git and CI/CD, enabling one-click recovery during outages.
- Tuned Linux kernel parameters and performed TCP congestion analysis, reducing in-game packet loss by 28% globally.
- Acted as technical lead for a 7-member engineering team, leading sprint planning, code reviews, and mentoring engineers on scalable backend design.
- Interviewed 50+ engineering candidates, raising team bar and improving technical hiring velocity.

Rackspace - Software Developer

Jun 2014 - May 2017

- Built and maintained backend services and cloud automation for Rackspace Public Cloud and OpenStack integrations.
- Contributed production-grade Python code to OpenStack components (Nova, Glance, Neutron), improving image management and compute orchestration.

- Built full-stack migration tool in Python, Redis, and Flask for Rackspace customers to move workloads to AWS — reducing migration time from days to hours.
- Implemented RESTful APIs for service provisioning, authentication, and metrics reporting, backed by DynamoDB and ElastiCache.
- Developed automated Linux image builders using Kickstart, KVM, Cloud-Init, and Ansible, ensuring consistent base image security and stability.
- Integrated continuous deployment pipelines using Jenkins + Terraform, reducing release cycles from biweekly to daily.

Deutsche Bank - Application Developer

Jun 2013 - Aug 2013

- Developed Java-based data visualization tools using Google Web Toolkit, improving analytics efficiency across finance teams.

Bloomberg LP - R&D Intern

May 2012 - Aug 2012

- Implemented multi-threaded C++ modules for internal analytics tools, optimizing computation by 40% through algorithmic improvements and memory profiling.

EDUCATION

Virginia Tech | 2011

B.S. in Computer Science